Bridges

Whether crossing a small gap or spanning more than 60 miles, a bridge must be able to withstand and transfer the forces of tension and compression from the bridge's weight and its load.

Types of bridges

While bridges come in all shapes and sizes, nearly all are variations on a few basic types. Beam and truss bridges are the simplest forms. Similar to laying a plank of wood between two banks, they can be used for only relatively short spans. The arch bridge is also best suited to shorter spans, unless multiple arches are joined together. Cable-stayed and, particularly, suspension designs offer the greatest scope for lengthy spans.

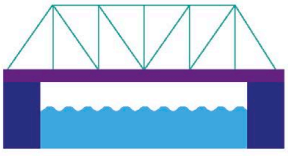
Beam bridge
In a beam bridge, piers or posts on either end support a flat deck. The deck consists of beams, such as hollow steel box girders.



Arch bridge
An arch constructed below the bridge supports the deck, transferring the compression forces to the piers.



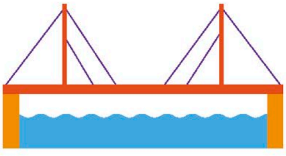
Truss bridge
In a truss bridge, the deck has added support from a girder framework with diagonal posts to counter compressive forces.



Cantilever bridge
This incorporates two "seesaws" whose ends meet in the middle. The ends are anchored on both sides.



Cable-stayed bridge
The deck is supported by multiple cables that are directly connected to one or more vertical towers.



Suspension bridges

In a cable-stayed bridge (see left), cables connect the deck directly to vertical towers. In a suspension bridge, the main cables connect the tops of the towers to anchor blocks embedded in banks at the ends of the bridge. The deck is supported by vertical suspension cables that hang from the main cables. This is a system that allows for very large spans.

Suspension bridge structure
The weight of the bridge's deck, and any extra load applied, is transmitted through suspension cables to the main cables, putting the suspension cables and main cables under tension. The main cables transmit the load to the fixed anchor blocks and to the towers. This produces a compressive force in the towers. The towers ultimately pass the load down to the foundations.

